

KUGLUKTUK, NU, Canada

WMO: 713320

Lat: 67.817N

Lon: 115.135W

Elev: 23

StdP: 101.05

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-39.7	-37.9	-42.4	0.1	-37.6	-40.7	0.1	-36.4	14.4	-17.9	13.1	-18.8	2.8	230	0.968

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	22.9	15.1	19.8	13.9	17.1	12.7	15.8	21.4	14.4	19.2	13.0	16.8	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	9.5	17.8	11.8	8.7	16.5	10.5	7.9	15.0	44.4	21.5	40.4	19.4	36.7	16.8	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.4	9.1	DB	-41.9	27.9	2.0	2.2	-43.3	29.4	-44.5	30.7	-45.6	31.9	-47.0	33.5	(4)
(5)			WB	-40.3	18.2	1.6	1.4	-41.5	19.1	-42.4	19.9	-43.3	20.7	-44.5	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.5	-26.5	-26.7	-24.7	-16.2	-4.7	5.7	11.1	9.5	3.8	-5.7	-16.7	-23.9		
	DBStd	15.09	6.71	6.66	6.75	6.80	5.77	4.44	3.89	3.91	3.99	5.11	5.97	6.53		
	HDD10.0	7243	1131	1028	1076	787	456	146	32	57	191	488	802	1051		
	HDD18.3	10162	1389	1262	1334	1037	714	381	227	276	436	746	1052	1309		
	CDD10.0	126	0	0	0	0	0	16	66	41	4	0	0	0		
	CDD18.3	4	0	0	0	0	0	1	2	2	0	0	0	0		
	CDH23.3	67	0	0	0	0	0	13	32	22	0	0	0	0		
	CDH26.7	11	0	0	0	0	0	3	5	3	0	0	0	0		
(14) Wind	WSAvg	4.3	5.0	4.7	4.2	3.7	3.7	3.7	3.8	4.0	4.5	4.5	4.7	4.6		
(15) Precipitation	PrecAvg	235	10	7	9	13	13	16	35	41	34	24	12	9		
	PrecMax	355	32	26	16	61	45	46	75	133	105	61	27	29		
	PrecMin	104	0	1	0	1	0	2	3	8	2	1	1	0		
	PrecStd	72	8	6	5	13	11	13	19	28	24	16	8	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.1	-8.1	-3.9	2.9	11.9	25.3	26.9	26.4	18.2	6.9	-0.8	-5.8		
		MCWB	-5.8	-8.8	-4.7	1.3	7.5	14.7	16.5	16.5	12.5	5.1	-1.4	-6.7		
	2%	DB	-10.4	-11.3	-7.7	0.1	7.6	19.9	23.3	22.2	14.4	3.1	-3.6	-8.7		
		MCWB	-11.0	-11.8	-8.4	-0.8	4.8	12.4	15.4	15.2	10.4	2.1	-4.3	-9.4		
	5%	DB	-14.0	-14.2	-11.2	-3.1	4.6	16.5	20.4	18.7	11.8	1.9	-5.8	-11.5		
		MCWB	-14.5	-14.7	-11.7	-4.0	2.7	11.4	14.5	13.8	8.9	1.1	-6.4	-11.9		
	10%	DB	-16.8	-16.5	-14.6	-6.1	2.7	13.0	17.7	15.7	9.5	0.6	-8.3	-14.2		
		MCWB	-17.2	-16.9	-15.0	-6.8	1.4	9.3	13.2	12.2	7.5	0.0	-8.9	-14.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.8	-8.6	-4.7	1.4	7.6	15.1	17.9	17.4	13.3	5.1	-1.5	-6.5		
		MCDB	-5.2	-8.1	-3.9	2.8	11.7	22.9	24.5	23.7	16.9	7.0	-0.9	-5.8		
	2%	WB	-10.8	-11.7	-8.4	-1.0	4.8	13.3	16.2	15.7	10.7	2.4	-4.3	-9.3		
		MCDB	-10.3	-11.2	-7.7	0.2	7.4	19.1	21.7	21.3	14.0	3.1	-3.7	-8.8		
	5%	WB	-14.3	-14.5	-11.6	-3.9	2.8	11.3	14.9	14.0	9.0	1.2	-6.4	-11.9		
		MCDB	-13.8	-14.1	-11.1	-3.1	4.4	16.1	19.6	18.1	11.2	1.9	-5.9	-11.5		
	10%	WB	-17.1	-16.8	-15.0	-6.8	1.5	9.3	13.6	12.4	7.7	0.0	-8.8	-14.6		
		MCDB	-16.7	-16.4	-14.6	-6.1	2.6	13.0	17.5	15.7	9.4	0.6	-8.3	-14.1		
(35) Mean Daily Temperature Range	MDBR		7.1	7.2	7.6	8.2	6.9	7.8	8.3	7.3	5.6	5.3	6.7	7.1		
	5% DB	MCDBR	9.4	9.6	9.9	9.6	9.6	14.9	13.3	12.9	10.4	4.8	7.8	9.3		
		MCWBR	9.2	9.4	9.7	9.0	7.3	9.0	7.1	7.2	6.8	4.3	7.6	9.1		
	5% WB	MCDBR	9.4	9.5	9.8	9.1	9.4	14.1	12.0	11.8	9.4	4.8	7.7	9.3		
		MCWBR	9.2	9.4	9.6	8.6	7.2	8.9	7.1	6.9	6.3	4.3	7.5	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.161	0.228	0.253	0.297	0.335	0.347	0.364	0.340	0.296	0.247	0.165	N/A		
	taud		1.894	2.060	2.207	2.143	2.177	2.339	2.368	2.467	2.529	2.210	1.914	N/A		
	Ebn at Noon		154	586	802	849	851	853	822	808	768	557	128	0		
	Edh at Noon		9	48	76	109	119	105	100	81	58	41	8	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.57	2.17	4.58	6.21	5.87	5.21	3.50	1.77	0.54	0.10	0.00		
	RadStd		0.00	0.04	0.13	0.21	0.42	0.37	0.38	0.41	0.21	0.08	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page